

IT6 Final Drill

Enhancing a Flask Web Application with REST API Functionality

Total Points: 20

Instructions:

Students will enhance an existing Flask web application by adding a meaningful REST API feature that implements full CRUD (Create, Read, Update, Delete) functionality. The feature must align with and improve the original application's purpose.

Grading rubric (and requirements):

1. Project Selection (2 points)

- Choose an existing Flask web application from GitHub, GitLab, or a similar platform. The project must:
 - Have clear documentation and a modular structure.
 - Be at least one year old (to ensure it's not a project created for this activity).
 - Not have been created by you.
 - Fork the repository and proceed.

2. Feature Implementation (8 points)

- Identify a useful feature that would benefit from REST API implementation.
- Implement complete CRUD operations for the chosen feature.
- Ensure proper error handling and use appropriate HTTP status codes (e.g., 200, 201, 400, 404).
- Maintain clean code standards and include proper documentation.
- Create a new branch for all your changes to make them easy to review.

Note: Only the API is required. No HTML, CSS, or JavaScript code is needed and will not be evaluated.

3. Testing (5 points)

- Write unit tests with 100% coverage for the new feature
- Include both positive and negative test cases
- Use the pytest or unittest framework
- Tests must verify all CRUD operations

4. Video Presentation (5 points)

- Deliver a 5-7 minute technical presentation that covers:
 - The original application's purpose
 - Your added feature and its value
 - Implementation details (routes, models, etc.)
 - Testing approach
- You must appear on camera during the presentation.

Technical Requirements:

1. API Design:

- Follow RESTful principles with proper resource naming and HTTP methods.
- Use JSON for all request and response formats.
- Include appropriate HTTP status codes.
- Provide API documentation using Swagger/OpenAPI (preferred).

2. Submission:

- Provide the original Repository
- Provide the URL of the cloned GitHub Repository with your changes (I suggest to make your new feature a separate branch, and so that I can easily see your changes from the code of the original author)
- README.md explaining your changes
- YouTube/Google-video direct link to your presentation (MP4 format only)

Important Note: Plagiarism will result in zero points. All work must be original except for the base application you're enhancing. Video presentation is mandatory; submissions without a video will not be graded.